

Q1 correct

Q2 -2

Q3 -0 good 104041

Q4 -0

Q5 -2

Homework Assignment 1

Q6: 6/16 Domain: 4/4 Range: 0/6 Sketch: 2/6

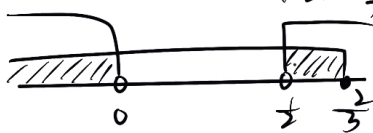
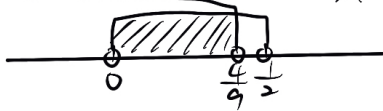
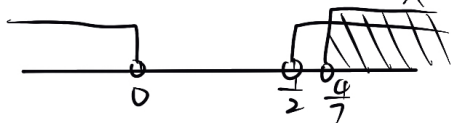
Missing the cases when $c = 0$ and $c < 0$.1. (14 points) Solve the inequality $\frac{1}{4} < \left| \frac{2x-1}{x} \right| \leq \frac{1}{2}$ in terms of intervals and illustrate the solution set on the real number line.

Q7 -3

First, to make $\frac{2x-1}{x}$ has meaning, $x \neq 0$

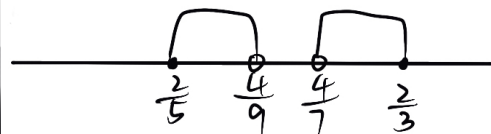
Then we can divide it by:

$$\begin{cases} \left| \frac{2x-1}{x} \right| \leq \frac{1}{2} & \textcircled{1} \\ \left| \frac{2x-1}{x} \right| > \frac{1}{4} & \textcircled{2} \end{cases}$$

In the function $f(x) = \frac{2x-1}{x}$, $f(x) > 0 \Rightarrow x > \frac{1}{2}$ we can know when $\begin{cases} 2x-1 > 0 \\ x > 0 \end{cases}$ or $\begin{cases} 2x-1 < 0 \\ x < 0 \end{cases}$,that is $x < 0$ or $x > \frac{1}{2}$, $f(x) > 0$ when $0 < x < \frac{1}{2}$, $f(x) < 0$ In ①, when $0 < x < \frac{1}{2}$, $\frac{1-2x}{x} \leq \frac{1}{2}$, $x \geq \frac{2}{5}$ when $x < 0$ or $x > \frac{1}{2}$, $\frac{2x-1}{x} \leq \frac{1}{2}$, $x \leq \frac{2}{3}$ In ②, when $0 < x < \frac{1}{2}$, $\frac{1-2x}{x} > \frac{1}{4}$, $x < \frac{4}{9}$ when $x < 0$ or $x > \frac{1}{2}$, $\frac{2x-1}{x} > \frac{1}{4}$, $x > \frac{4}{7}$ Because $\frac{2}{5} < \frac{4}{9}$, $\frac{2}{3} > \frac{4}{7}$

In conclusion,

$$x \in \left[\frac{2}{5}, \frac{4}{9} \right) \cup \left(\frac{4}{7}, \frac{2}{3} \right]$$



2. (14 points) Sketch the set $\{(x, y) \in \mathbb{R}^2 : \sqrt{2x^2 + 16x + 34} + 2 \leq y \leq -|x + 4| + 5\}$ in the xy -plane.

We can set two functions,

$$f(x_1) = -|x+4| + 5$$

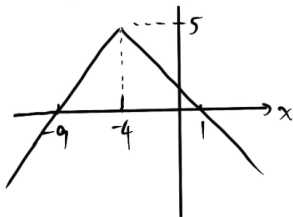
$$f(x_2) = \sqrt{2x^2 + 16x + 34} + 2$$

In the $f(x_1)$, we can find the domain is $\mathbb{R}$.

$$-|x+4| + 5 \iff -|x+4| \iff -|x| \leq |x|$$

that means to get $f(x_1)$, we should firstly reflect $|x|$ about the x axis, then move it horizontally to the left with 4 units, finally move it vertically up with 5 units.

The graph of $f(x_1)$ is:



In the $f(x_2)$, we can find the domain is

$$2x^2 + 16x + 34 \geq 0, \text{ since } \Delta = (16)^2 - 4 \times 2 \times 34 = -16$$

that means $\forall x \in \mathbb{R}, f(x_2) \geq 0$

So the domain is $\mathbb{R}$.

Then we easily find $f(x_2) \neq f(-x_2), f(x_2) \neq -f(-x_2)$

So $f(x_2)$ is not odd or even.

In addition, we should find the increasing and decreasing intervals.

Assume $g(x) = 2x^2 + 16x + 34$, then $f(x_2) = \sqrt{g(x)} + 2$

Since "2" don't affect monotonicity, we can focus on $\sqrt{x}$, which is increasing function.

$$g(x) = 2x^2 + 16x + 34 = 2(x+4)^2 + 2$$

Then we can know $(-\infty, -4)$ is decreasing

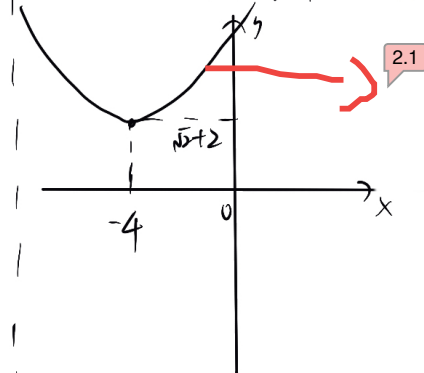
$[-4, \infty)$ is increasing

Also, we should find minimum and maximum

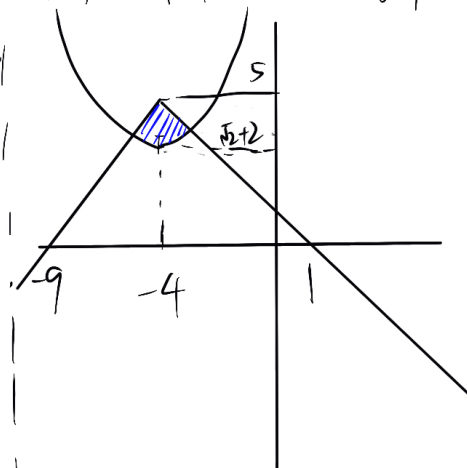
We can find they are no maximum

$$f(x_2)_{\min} = f(-4) = \sqrt{2} + 2$$

Then we can get graph of $f(x_2)$



Then, we put these two graph together.



The blue area is the answer.

3. (14 points) Recall that we use the symbol $\parallel$ when two lines are parallel or the same, and the symbol $\perp$ when two lines are perpendicular. Let l, r, s be non-vertical lines. Show that:

- (a) If $l \parallel r$ and $r \parallel s$, then $l \parallel s$.
- (b) If $l \parallel r$ and $r \perp s$, then $l \perp s$.
- (c) If $l \perp r$ and $r \perp s$, then $l \parallel s$.

Assume that l, r, s lines' slopes respectively are $k_l, k_r, k_s (\neq 0)$

(a) $l \parallel r$ means $k_l = k_r$, $r \parallel s$ means $k_r = k_s$

since $k_l = k_r, k_r = k_s \Rightarrow k_l = k_s$, $l \parallel r$ and $r \parallel s \Rightarrow l \parallel s$

(b) $l \parallel r$ means $k_l = k_r$, $r \perp s$ means $k_r \cdot k_s = -1$

$$k_l = k_r, k_r \cdot k_s = -1 \Rightarrow k_l \cdot k_s = -1$$

so if $l \parallel r$ and $r \perp s$, then $l \perp s$

(c) $l \perp r$ means $k_l \cdot k_r = -1$, $r \perp s$ means $k_r \cdot k_s = -1$

$$k_l \cdot k_r = -1, k_s \cdot k_r = -1, k_r \neq 0 \Rightarrow k_l = k_s$$

so if $l \perp r$ and $r \perp s$, then $l \parallel s$

4. (14 points) Find the line perpendicular to the line $2x - \sqrt{5}y = 6 + 4\sqrt{5}$ that passes through the intersection of the conics $4x^2 + 5y^2 - 24x + 40y + 96 = 0$ and $\frac{(y+4)^2}{4} - \frac{(x-3)^2}{5} = 1$, which is the closest to the origin.

We can find the line $2x - \sqrt{5}y = 6 + 4\sqrt{5}$ slope is $k_1 = \frac{2}{\sqrt{5}}$

Assume that the line $l_2: y = k_2x + b_2$ is what we get

According to question, $k_1 \cdot k_2 = -1$, then we get $k_2 = -\frac{\sqrt{5}}{2} = -\frac{A}{B}$

Then we can disposal the conics

$$4x^2 + 5y^2 - 24x + 40y + 96 = 0$$

$$4(x^2 - 6x) + 5(y^2 + 8y) + 96 = 0$$

$$4(x-3)^2 + 5(y+4)^2 = 20$$

$$\frac{(x-3)^2}{5} + \frac{(y+4)^2}{4} = 1$$

$$\begin{cases} \frac{(x-3)^2}{5} + \frac{(y+4)^2}{4} = 1 \\ \frac{(y+4)^2}{4} - \frac{(x-3)^2}{5} = 1 \end{cases} \Rightarrow \begin{cases} x=3 \\ y=-2 \end{cases} \text{ or } \begin{cases} x=3 \\ y=-6 \end{cases}$$

Since we need the closest to the origin.

$$|(3, -2)| = \sqrt{3^2 + (-2)^2} < \sqrt{3^2 + (-6)^2} = |(3, -6)|$$

The intersection of the conics is $(3, -2)$

$$\text{Then we get } -2 = -\frac{\sqrt{5}}{2} \cdot 3 + b_2$$

$$b_2 = \frac{3\sqrt{5}}{2} - 2$$

$$\text{In conclusion, the line } l_2: y = -\frac{\sqrt{5}}{2}x + \frac{3\sqrt{5}}{2} - 2$$



5. (14 points) Find all real numbers t such that $\cos t = \sin^2 t - \frac{1}{4}$.

$$\cos t = \sin^2 t - \frac{1}{4}$$

$$\cos t = 1 - \cos^2 t - \frac{1}{4}$$

$$\cos^2 t + \cos t - \frac{3}{4} = 0$$

$$4\cos^2 t + 4\cos t - 3 = 0$$

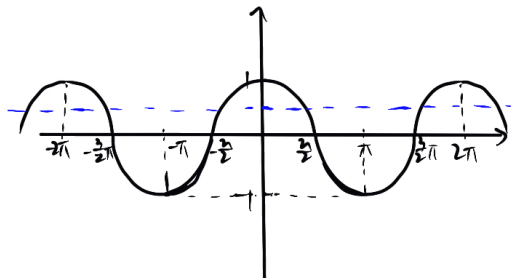
$$(2\cos t + 3)(2\cos t - 1) = 0$$

$$\cos t = -\frac{3}{2} \text{ or } \frac{1}{2}$$

$$\text{since } \cos t \in [-1, 1]$$

$$\cos t = \frac{1}{2}$$

The graph of $\cos t$ is:



In this graph, we can find that

$$\cos t = \frac{1}{2} \Leftrightarrow t = \pm \frac{\pi}{3} + 2\pi k, k \in \mathbb{Z}$$

5.1

6. (16 points) Let c be a fixed real number. Find the domain, the range, and sketch the following function.

$$f(x) = \begin{cases} c \left(\frac{x^3 + 3x^2 + 3x + 1}{x+1} \right) & \text{if } x < -\frac{\pi}{6}, \quad \textcircled{1} \\ 2 \cos x - \sqrt{3} & \text{if } -\frac{\pi}{6} \leq x \leq 3\pi, \quad \textcircled{2} \\ \frac{x^2 - 6\pi x + 9\pi^2}{x - 3\pi} - 2 - \sqrt{3} & \text{if } 3\pi < x < 5\pi, \quad \textcircled{3} \\ e^{-\frac{x}{5\pi}} & \text{if } x \geq 5\pi. \quad \textcircled{4} \end{cases}$$

$$2 \times \cos\left(\frac{3\pi}{2}\right) - \sqrt{3} = 0$$

We can divide this function by four parts.

i) In part ①, we find $x+1 \neq 0$, that is $x \neq -1$. So the domain is

$$x \in (-\infty, -1) \cup (-1, -\frac{\pi}{6})$$

$$c \left(\frac{x^3 + 3x^2 + 3x + 1}{x+1} \right) = c \left(\frac{(x+1)^3}{x+1} \right) = c(x+1)^2$$

$x \in (-\infty, -1)$, it's decreasing

$x \in (-1, -\frac{\pi}{6})$, it's increasing

ii) In part ②, the domain is $-\frac{\pi}{6} \leq x \leq 3\pi$

$$-\frac{\pi}{6} \leq x \leq 3\pi \Rightarrow -1 \leq \cos x \leq 1 \Rightarrow -2 - \sqrt{3} \leq 2 \cos x - \sqrt{3} \leq 2 - \sqrt{3}$$

$$x \in [-\frac{\pi}{6}, 0], \text{ it's increasing}$$

$$x \in (0, \pi) \cup (2\pi, 3\pi], \text{ it's decreasing}$$

iii) In part ③, the domain is $(3\pi, 5\pi)$

$$\frac{x^2 - 6\pi x + 9\pi^2}{x - 3\pi} - 2 - \sqrt{3}$$

$$= \frac{(x - 3\pi)^2}{x - 3\pi} - 2 - \sqrt{3}$$

$$= x - 3\pi - 2 - \sqrt{3}$$

$$3\pi < x < 5\pi \Rightarrow -2 - \sqrt{3} < x - 3\pi - 2 - \sqrt{3} < 2\pi - 2 - \sqrt{3}$$

since " $-2 - \sqrt{3}$ " don't affect monotonicity

$f(x) = x$ is increasing function

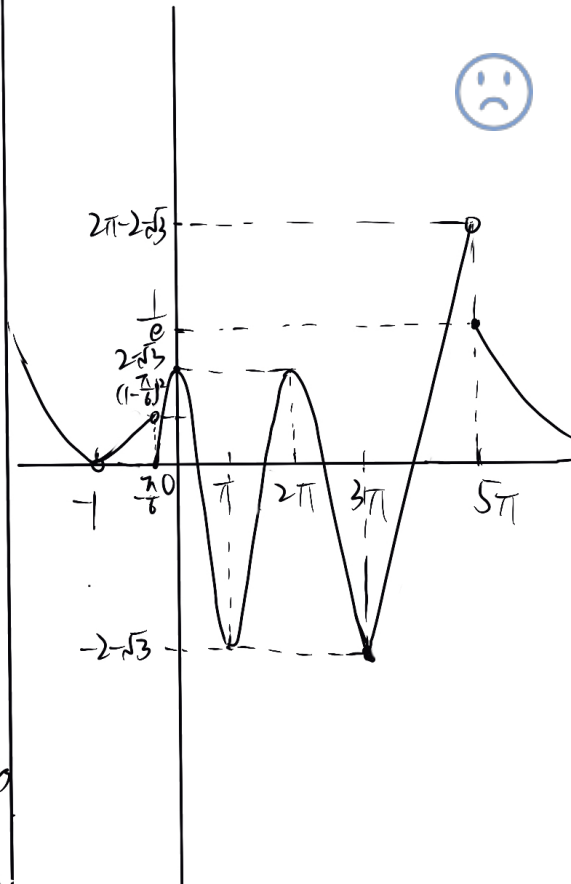
$x - 3\pi - 2 - \sqrt{3}$ is increasing

iv) In part ④, the domain is $x \geq 5\pi$

$$x \geq 5\pi \Rightarrow -\frac{x}{5\pi} \leq -1 \Rightarrow 0 < e^{-\frac{x}{5\pi}} \leq \frac{1}{e}$$

since $f(x) = e^x$ is increasing, $f(x) = -x$ is decreasing

$e^{-\frac{x}{5\pi}}$ is decreasing function



In conclusion, the domain is $(-\infty, -1) \cup (-1, \infty)$
the range is

7. (14 points)

(a) Show that, if $0 < a < b$ and $0 < c < d$ then $ac < bd$.

(b) Using part a), show that if $0 < x < y$ then $x^3 < y^3$.

(a) since $0 < a < b$, $0 < c < d$
when we multiply ^{them} the original
direction is unchanged. 7.1

So we get $ac < bd$

(b) $0 < x < y \Rightarrow \underline{0 < x^2 < y^2}$ **PROVE**

According to (a), $0 < x < y$ and $0 < x^2 < y^2$
then $x \cdot x^2 < y \cdot y^2$
that is $x^3 < y^3$



Index of comments

- 2.1 By simplification, a hyperbola can be obtained.
- 5.1 These are not the solutions of the cosine.
- 6.1 The range is what? Please finish your conclusion next time.
- 7.1 more process to prove.
Since $0 < a < b, 0 < c < d$, then $0 < ac < bc, 0 < bc < bd$, then $ac < bd$.